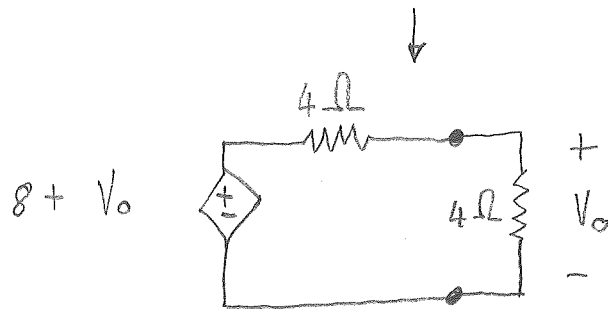
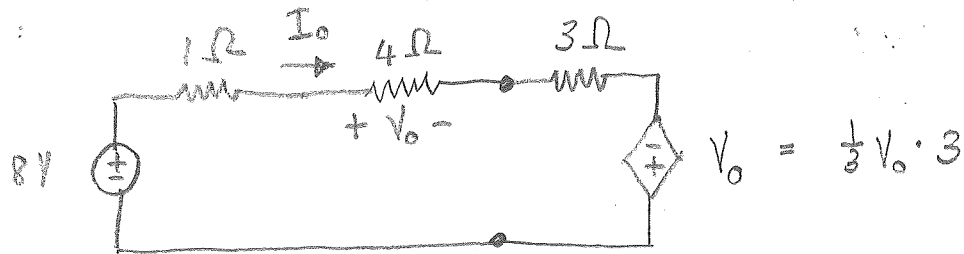


EBN 122 2022
SEMESTER TEST 2
MEMORANDUM

01-03:



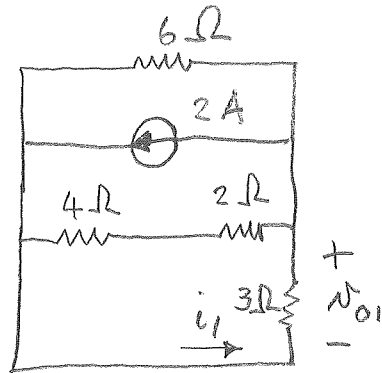
01. $R_x = 4\Omega$

02. $A = 1$

03. $B = 8V$

04 - 05 :

04 :



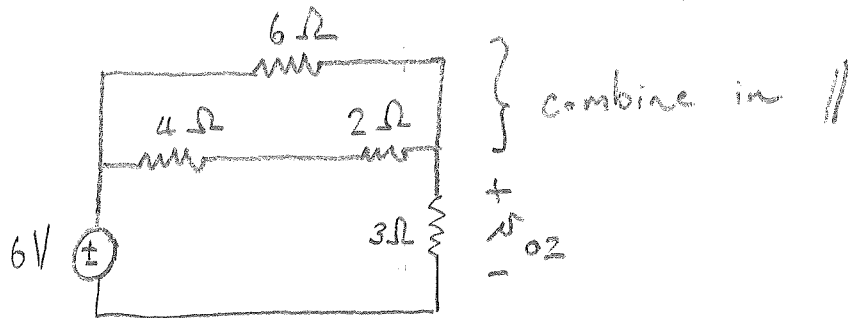
$$(4+2) \parallel 6 = 3\Omega$$

current div :

$$i_1 = 2 \cdot \frac{3}{3+3} = 1\text{ A}$$

$$v_{01} = -3i_1 = -3\text{ V}$$

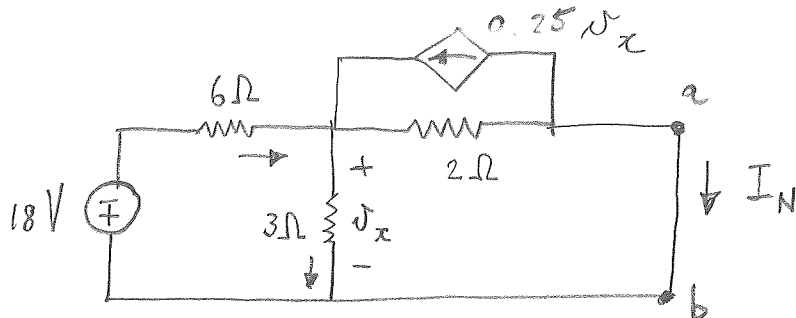
05 :



$$v_{02} = 6 \cdot \frac{3}{3+3} = 3\text{ V}$$

06-08:

06.



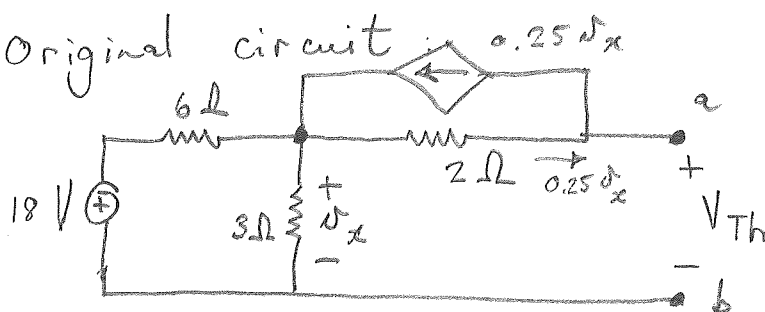
$$\text{KCL: } \frac{-18 - v_x}{6} + 0.25v_x = \frac{v_x}{3} + \frac{v_x}{2}$$

$$-18 - v_x + 1.5v_x = 2v_x + 3v_x$$

$$-18 = 4.5v_x \therefore \underline{v_x = -4 \text{ V}}$$

$$\begin{aligned} I_N &= \frac{v_x}{2} - 0.25v_x \\ &= -1 \text{ A} \end{aligned}$$

0.7 Original circuit:



$$v_x = -18 \frac{3}{6+3} = -6 \text{ V}$$

$$\begin{aligned} V_{Th} &= v_x - 2(0.25v_x) = -6 - 2(-1.5) \\ &= -3 \text{ V} \end{aligned}$$

$$08. \quad R_{Th} = \frac{V_{Th}}{I_N} = \frac{-3}{-1} = 3 \Omega \quad \leftarrow \text{continuity marking.}$$

09-10

$$\begin{aligned} 09 \quad R_{Th} &= (2 + 3 + 4) \parallel 5 + 2 \\ &= \frac{9 \cdot 5}{9 + 5} + 2 = 5.21 \, \Omega = R_L \end{aligned}$$

→

$$\begin{aligned} 10 \quad V_{Th} &= 24 \cdot \frac{5}{2 + 3 + 4 + 5} \\ &= 8.57 \, V \end{aligned}$$

$$P_{max} = \frac{V_{Th}^2}{4 R_{Th}} = 3.36 \, W \quad \leftarrow \text{continuity marking}$$

$$\begin{aligned} \underline{11} \quad i_o^{new} &= i_o \cdot \frac{27}{20} \\ &= -4 \cdot \frac{27}{20} = -5.4 \, A \end{aligned}$$

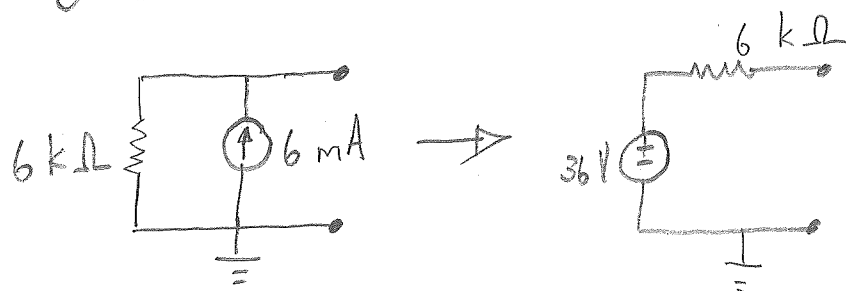
12. Non-inverting amplifier:

$$A_o = 144 = \left(1 + \frac{5000}{R_1}\right) 24$$

$$6 = 1 + \frac{5000}{R_1} \therefore R_1 = 1 \text{ k}\Omega$$

13: Inverting amplifier:

Source transformation:



$$V_o = - \frac{12000}{6000} 36 = -72 \text{ V}$$

14-15 : Inverting amplifier:

$$\underline{14:} \quad V_1 = - \frac{20000}{10000} 8$$

$$= -16 \text{ V} \text{ but op-amp saturates at } -12 \text{ V}$$

$$\therefore V_1 = -12 \text{ V} \rightarrow$$

$$15: \quad i_1 = \frac{0 - V_1}{20000} = +0.6 \text{ mA} \rightarrow$$

continuity
marking

16-17

Inverting amplifier in cascade with
summing amplifier

$$\underline{16}: \quad v_x = - \frac{60\,000}{30\,000} (-0.4) = 0.8 \text{ V}$$

$$\begin{aligned} 17: \quad v_y &= - \left(\frac{40}{10} (-0.2) + \frac{40}{20} v_x \right) \\ &= -0.8 \text{ V} \end{aligned}$$

continuity
marking

18-20

18

KCL at a:

continuity
marking

$$\frac{2 - V_a}{1} = \frac{V_a}{3} + \frac{V_a}{5}$$

$$10 - 5V_a = \frac{5}{3}V_a + V_a$$

$$10 = V_a \left(5 + \frac{5}{3} + 1\right) \therefore V_a = 1.304 \text{ V}$$

19

KVL: $V_b = V_a = 1.304 \text{ V}$

20

KCL:

$$\frac{0 - V_b}{2} + \frac{0 - V_a}{5} + \frac{0 - V_c}{10} = 0$$

$$-5V_b - 2V_a - V_c = 0$$

$$V_c = -2V_a - 5V_b$$

$$= -2(1.304) - 5(1.304)$$

$$= -9.128 \text{ V}$$

21-22

21

$$\frac{6 - (-3)}{10} = \frac{-3 - V_1}{20}$$

continuity
marking

$$18 = -3 - V_1 \therefore V_1 = -21 \text{ V}$$

22

$$\frac{-3 - V_1}{20000} = i_1 + \frac{V_1}{30000}$$

$$-4.5 - 1.5V_1 = 30000 i_1 + V_1$$

$$30000 i_1 = -4.5 - 2.5V_1 \therefore i_1 = 1.6 \text{ mA}$$

23

$C_{eq} : 6 \parallel 6 \parallel (3\text{mF in series with } 3\text{mF})$

$$C_{eq} = 6 + 6 + \frac{3.3}{3+3}$$
$$= 13.5 \text{ mF}$$

24-25

24 Voltage across $\parallel$ combination

$$v_1(t) = 3 \frac{di_1}{dt} = -9e^{-3t} \text{ A}$$

$$i_2(t) = \frac{1}{6} \int_0^t v_1(t) dt + i_2(0)$$
$$= \frac{1}{6} (-9) \left(-\frac{1}{3}\right) e^{-3t} \Big|_0^t + 2$$
$$= 0.5e^{-3t} - 0.5 + 2$$
$$= 0.5e^{-3t} + 1.5 \text{ A}$$

$$i_2(t=1s) = 1.53 \text{ A}$$


25

$$i(t) = i_1(t) + i_2(t)$$

$$= e^{-3t} + 0.5e^{-3t} + 1.5 = 1.5e^{-3t} + 1.5 \text{ A}$$

$$v_o(t) = 2 \frac{di}{dt} = -9e^{-3t} \text{ V}$$

$$\therefore v_o(t=1s) = -0.448 \text{ V}$$
